A-A-1897
July 24, 1981

COMMERCIAL ITEM DESCRIPTION

CASSETTE, MAGNETIC TAPE, 3/4-INCH, U-MATIC, VIDEO COLOR

The General Services Administration has authorized the use of this commercial item description.

This commercial item description covers new 3/4-inch color video tape cassettes for recording use on U-Matic type video recorders/reproducers. All cassettes having playing time capabilities of 10, 15, 20, 30, 40, 50, and 60 minutes are covered.

Salient characteristics:

Dimensions. The cassettes shall conform to the dimensional requirements of ANSI C98.22.

Magnetic tape width. The width of the magnetic tape shall be 0.748 inch ± 0.001 inch.

Yield strength. The 1 percent offset yield point, the 3 percent elongation point and the breaking point of the tape shall equal or exceed a force per unit area of 13,000 pounds per square inch.

Dropouts. The average number of video dropouts per minute and the maximum number of video dropouts in any one minute shall not exceed 20 min. All discontinuities in RF output, which for a period of 10 microseconds or more are equal to or exceed an 18 db decrease in average RF output level shall be counted as dropouts.

Audio uniformity. The uniformity of the signal output at test frequencies of 1,000 Hz and 6,000 Hz, when tested as specified below, shall not exceed 3.0 db.

Short term uniformity. The frequencies specified above shall be recorded at standard audio record level and with operating bias current over the entire tape length. On playback, the output voltage variations shall be measured by means of a linear recording oscillograph having a response which is uniform to at least 200 Hz. The tape uniformity shall be measured by determining the maximum variation in output level over any 10 second interval, with all nonrecurring pips being disregarded.

Long term uniformity. The test procedure shall be similar to that specified above. The tape uniformity shall be measured by determining the variation of the peak output over the tape length.

Operational life. The cassette shall not exhibit any deterioration in functional performance after subjection to the operational tests specified herein. An optinum video recording of standard 75 percent NTSC color bars shall be made over a 1 minute length of tape at the beginning of the cassette. The recording shall then be reproduced to verify satisfactory picture quality. The cassette shall then be subject to 10 fast forward and rewind cycles over the entire tape length. The recorded section of tape shall then be reproduced a total of 25 times and the picture quality evaluated. A final re-recording and reply of the initially recorded section shall be made. After completion of all tests the cassette shall have shown no evidence of picture degradation, tape sticking, excessive oxide shed, head clogging, etc.

Workmanship. The cassettes, magnetic tape, and containers shall be manufactured and processed in a careful and workmanlike manner in accordance with good practice. The entire length of the recording tape shall be continuous and splice free, and both the oxide surface and base shall be free from any raised areas, dust, flakes, powder, holes, foreign matter, scratches, creases, and other defects which would render the tape unsuitable for its intended use.

Toxic compounds. The tape shall not produce toxic effects through inhalation, ingestion, contact with the skin in normal Government usage.

Quality assurance. All lots offered for delivery must meet the requirements specified herein. The Government may from time to time require inspection of lots offered for delivery in accordance with MIL-STD-105, Special Inspection Level S-2, at an acceptable quality level of 10 percent defective. Failure of samples to comply would result in rejection of material and be cause for further lot by lot inspection.

Certification. The contractor shall certify that the product offered meets the salient characterstics of this description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product sold in the commercial marketplace. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Regulatory requirements. The manufacturer shall utilize recovered materials to the maximum extent practicable.

Packaging, packing, and marking. Packaging, packing, and marking shall be as specified in the contract or order.

NOTES: Definition of terms

Reference tape. A cassette containing an unrecorded length of tape (1,175 feet minimum) of the same type the magnetic tape manufacturer wishes to supply, shall be furnished to the General Services Administration for certification that the tape will meet the system performance specifications (including video and audio frequency response, video and audio signal-to-noise and audio distortion) of the reference test recorder. When found acceptable, this tape will become the reference tape for the manufacturer's particular product type and it will be used for establishing the operating levels and conditions for the reference test recorder when submitted samples of the particular tape type are evaluated for conformance to the manufacturer's specifications and when tested as specified herein.

Reference test recorder. The reference test recorder shall be a 3/4-inch U-Matic video cassette recorder/reproducer which is capable of recording video, audio, and tracking control signals in accordance with the standard U-Matic recording format. For the tests specified herein the General Services Administration will use Sony 2600, Sony 2800 or equivalent video recording-reproducing systems.

Operating audio bias current. Operating bias current is that bias current through the recording head which will give peak output from the reference tape.

Maximum audio record level. The maximum audio record level is that input level of a 333 Hz sinusodial signal recorded on the reference tape at operating bias current such that on playback the output signal will have 3 percent third harmonic distortion.

Standard audio record level. The standard audio record level is 8 db below the maximum audio record level.

ANSI Standards are available from American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.

Copies of MIL-STD-105 are available from Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

MILITARY INTEREST

Military Coordinating Activity:	Preparing Activity:
Army - CR	GSA-ADTS

GSA DC-01105238

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